# Yuk Tung Samuel Fang

Also published as **Yudong Fang**

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## Research Interests

Embodied AI and robotics, particularly semantic mapping, persistent world representations, embodied memory, and continual adaptation for long-term interaction.

## Education

**Nanjing University** Suzhou, China

B.Sc. in Intelligence Science and Technology Expected Jun. 2027

- **GPA:** 4.48/5.00
- **Selected coursework:** Database (96), Data Structures and Algorithms (95), Data Mining (95), Operating Systems (95), Programming Training (94.9), Machine Learning (94), Deep Learning (94)

## Publications

- **Y.T.S. Fang**, Z. Shi, J. Qiu, Z. Chen, J. Shi, H. Xu, J. Huo, Y. Gao. **INHerit-SG: Incremental Hierarchical Semantic Scene Graphs with RAG-Style Retrieval.** *ICRA 2026 Workshop on Robots Meet Prior Maps*; Oral Presentation; **Best Presentation Finalist**. [arXiv] [Project]
  *Role: First author and project lead.*
- H. Ding, Z. Xu, **Y.T.S. Fang**, Y. Wu, Z. Chen, J. Shi, J. Huo, Y. Zhang, Y. Gao. **LaViRA: Language-Vision-Robot Actions Translation for Zero-Shot Vision-Language Navigation in Continuous Environments.** *IEEE International Conference on Robotics and Automation (ICRA)*, 2026. [arXiv] [Project]
  *Role: Implemented the Unitree Go1 real-world robot setup and deployment; built an exploratory RAG prototype and contributed to visualization.*

## Selected Research Experience

**Incremental Hierarchical Semantic Scene Graphs (INHerit-SG)** Oct. 2025–Jul. 2026

*Inference and Learning Research Group, Nanjing University*

*Advised by Prof. Jieqi Shi, Prof. Hao Xu, and Prof. Jing Huo*

- Developed an incremental semantic scene representation organized as a Floor–Room–Area–Object hierarchy, with asynchronous geometric mapping and semantic reasoning.
- Designed a query-conditioned retrieval pipeline combining structured constraint parsing, candidate filtering, relational reasoning, and selective visual reranking for compositional and ambiguous queries.
- Built HM3DSem-SQR with 6,084 queries across 36 scenes and validated the system in simulation, real-world trajectories, and human studies, achieving 37.7% SR@1m, 70.6% semantic accuracy, and 60.0% real-world success.
- Following the workshop version, conducted controlled ablations and redesigned the pipeline around structured candidate recall and selective visual reranking, removing less effective hierarchy and multi-agent components. The revised system achieved 33.56% Strict Top-1, 62.27% Recall@5, and 37.62% SR@1m on Main-864, with best or tied-best results across all six query categories, and exceeded the best reported trained method on ViGiL3D by 5.1 pp F1.

**Life-Long Mobile Manipulation and Embodied Memory Evaluation** Mar. 2026–Present

*Nanjing University; Summer Intern at LINS Lab, National University of Singapore (Jul.–Aug. 2026)*

*Advised by Prof. Jieqi Shi and Prof. Lin Shao*

- Built a long-horizon mobile-manipulation and environment-perturbation platform in Habitat/ReplicaCAD and AI2-THOR/ProcTHOR, alternating task execution and memory updates under object relocation, hiding, state changes, disappearance, and spawning.
- Developed a three-level diagnostic benchmark to isolate memory-specific capabilities from downstream planning and execution: retrieval (L1), belief-state maintenance under partial observability (L2), and memory-conditioned action selection with fixed perception/control (L3).
- Built 2,966 L1 queries, 2,646 L2 plans, and 108 multi-goal tasks. DynaMem and Embodied VideoAgent achieved 81.2%/91.2% L1 retrieval accuracy; DynaMem reached 75.0%/83.3% on L2 state updates, while correct memory improved L3 goal completion from 3.3% to 10.6%.

**Zero-Shot Vision-Language Navigation (LaViRA)** May 2025–Sep. 2025
*Inference and Learning Research Group, Nanjing University*
*Advised by Prof. Jieqi Shi and Prof. Jing Huo*

- Built a reusable Unitree Go1 real-robot experimental platform and adapted LaViRA to it, including communication and sensor integration, navigation execution, deployment, and debugging.
- Built and tested an exploratory retrieval-augmented generation (RAG) prototype and contributed to the visualization of experiments and results.
- Contributed to the system achieving 38.3% Success Rate and 28.3% SPL on VLN-CE, outperforming InstructNav by 16.1 and 17.9 percentage points, respectively.

**Active Exploration with Semantic Map Prediction (SEA)** Sep. 2025–Oct. 2025
*Inference and Learning Research Group, Nanjing University*
*Advised by Prof. Jieqi Shi and Prof. Jing Huo*

- Explored active navigation using semantic map prediction to infer unobserved regions and guide subsequent exploration.
- Adapted SEA to the reusable Unitree Go1 real-robot platform developed during LaViRA, and conducted system integration, interface debugging, deployment, and real-robot experiments.

## Honors and Awards

- **First Prize**, NJU Scholarship for Hong Kong, Macao, and Overseas Chinese Students, Sophomore Year; **5 recipients university-wide**
- **Third Prize**, NJU Scholarship for Hong Kong, Macao, and Overseas Chinese Students, Freshman Year

## Academic Service

- Invited Reviewer, *IEEE Robotics and Automation Letters (RA-L)*, 2026

## Technical Skills

- **Programming and Tools:** Python, C++, PyTorch, OpenCV, Linux, Git, Docker, Conda
- **Robotics and Simulation:** ROS, Habitat/ReplicaCAD, AI2-THOR/ProcTHOR
- **Embodied AI and 3D Vision:** semantic mapping, 3D scene graphs, vision-language and large language model integration, retrieval-augmented generation
- **Robot Systems:** Unitree Go1, NVIDIA Jetson Orin, RGB-D sensing, navigation system integration, real-robot deployment and debugging, 3D modeling and printing